

MINI PROJECT REPORT ON
“SOLAR PANEL MAINTENANCE ALERT SYSTEM”

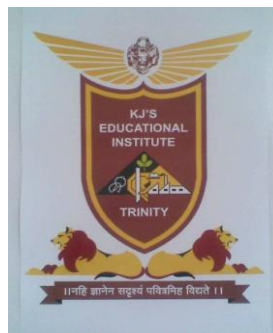
SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY,
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OF

THIRD OF ENGINEERING
(ELECTRONICS & TELECOMMUNICATION)

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DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING
(Accredited By NAAC 'A+' Grade)

PUNE - 411 048

2026-2027

CERTIFICATE

I hereby certify that the work which is being presented in the Mini Project report entitled “**SOLAR PANEL MAINTENANCE ALERT SYSTEM**” in partial fulfillment of the requirements for the award of **the Third Year Degree in Electronics & Telecommunication Engineering** and submitted to the Electronics & Communication Engineering Department of KJEI’s Trinity Academy of Engineering College, Pune is an authentic record of my own work carried out during a period from **January 2026 to May 2026** under the supervision of **Prof. [Supriya A. Dhoke], Mini Project Coordinator, E&TC Department.**

The matter presented in this project Report has not been submitted by me for the award of any other degree elsewhere. This is to certify that the above statements are correct to the best of my knowledge.

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ACKNOWLEDGEMENT

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We express our sincere gratitude to **Mrs. Supriya A. Dhoke**, Mini Project Coordinator (E&TC Dept.), for her support and valuable suggestions.

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We are thankful to all the teaching and non-teaching staff members of the **Electronics & Telecommunication Engineering** Department for their cooperation and support.

We would also like to thank the library staff for providing the necessary resources and information required for our project work.

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Date:

Place: TAE, Pune.

ABSTRACT

Solar energy is one of the most important renewable energy sources used for power generation. However, the efficiency of solar panels decreases due to dust accumulation, shading, and other environmental factors. Regular maintenance is necessary to ensure maximum output, but manual monitoring is time-consuming and not always reliable.

This project, titled “Solar Panel Maintenance Alert System,” aims to develop an efficient and low-cost solution to monitor the performance of a solar panel and provide alerts when maintenance is required. The system uses a voltage sensor to continuously measure the output of the solar panel. The measured data is processed using an ESP8266 microcontroller, which compares the actual output with a predefined threshold value.

If the output voltage drops below the set limit, the system identifies it as a possible fault condition such as dust accumulation or shading and generates an alert. The alert can be indicated through an LED or sent to a mobile device using IoT technology. A MOSFET is used to control the load, such as an LED strip, ensuring proper power management.

This system helps in reducing manual effort, improving the efficiency of solar panels, and ensuring timely maintenance. It is cost-effective, easy to implement, and suitable for applications such as home solar systems, agricultural setups, and small solar power installations.

The proposed system demonstrates how embedded systems and IoT can be effectively used to enhance renewable energy management and reliability.

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INTRODUCTION

In recent years, the demand for renewable energy sources has increased significantly due to the depletion of conventional energy resources and the growing concern for environmental protection. Solar energy is one of the most widely used renewable energy sources because it is clean, abundant, and sustainable. Solar panels are used to convert sunlight into electrical energy, which can be utilized for various applications such as home lighting, agriculture, and small-scale industries.

However, the efficiency of solar panels is highly dependent on environmental conditions. Factors such as dust accumulation, shading, bird droppings, and poor maintenance can reduce the output power of solar panels. Regular monitoring and maintenance are essential to ensure maximum efficiency, but manual inspection is not always practical, especially in remote or large-scale installations.

To overcome these challenges, this project proposes a “Solar Panel Maintenance Alert System” that can automatically monitor the performance of a solar panel and provide alerts when maintenance is required. The system uses a voltage sensing mechanism to continuously measure the output of the solar panel. The measured data is processed using an ESP8266 microcontroller, which compares it with a predefined threshold value.

If the output voltage falls below the desired level, the system identifies it as a fault condition and generates an alert through an LED indicator or mobile notification using IoT technology. This helps in taking timely action to clean or repair the panel, thereby improving its performance and lifespan.

The proposed system is cost-effective, easy to implement, and suitable for real-world applications such as residential solar systems, agricultural fields, and small solar power plants. It demonstrates the effective use of embedded systems and IoT in renewable energy management.

LITERATURE SURVEY

The increasing use of solar energy systems has led to the need for efficient monitoring and maintenance techniques to ensure maximum performance. Various research works and existing systems have focused on improving efficiency of solar panels by detecting faults and providing timely maintenance alerts [1].

Several studies have shown that dust accumulation and shading significantly reduce the efficiency of solar panels. Traditional methods of maintenance involve manual inspection, which is time-consuming, less accurate, and not suitable for large-scale installations. To overcome these limitations, automated monitoring systems have been developed using sensors and microcontrollers [2].

Many modern systems use voltage and current sensors to continuously monitor the output of solar panels. If any abnormal drop in output is detected, the system identifies it as a fault condition. Some advanced systems also use IoT technology to send real-time data and alerts to users through mobile applications or web platforms [3].

In this project, the concept of monitoring solar panel performance using a voltage sensor and microcontroller is adopted. The ESP8266 microcontroller is used due to its built-in Wi-Fi capability, which makes it suitable for IoT-based applications. The use of a MOSFET for load control further improves the system's efficiency and reliability [4].

The guidance and technical knowledge provided by **Prof. Dhage Sir** from our college played an important role in understanding the practical aspects of the system, component selection, and implementation of the project.

Based on the reviewed literature and guidance, the proposed system is designed to be simple, cost-effective, and efficient in providing maintenance alerts for solar panels. It helps in reducing manual effort and ensures better performance of the solar energy system.

PRESENT WORK

1. Circuit Diagram:-

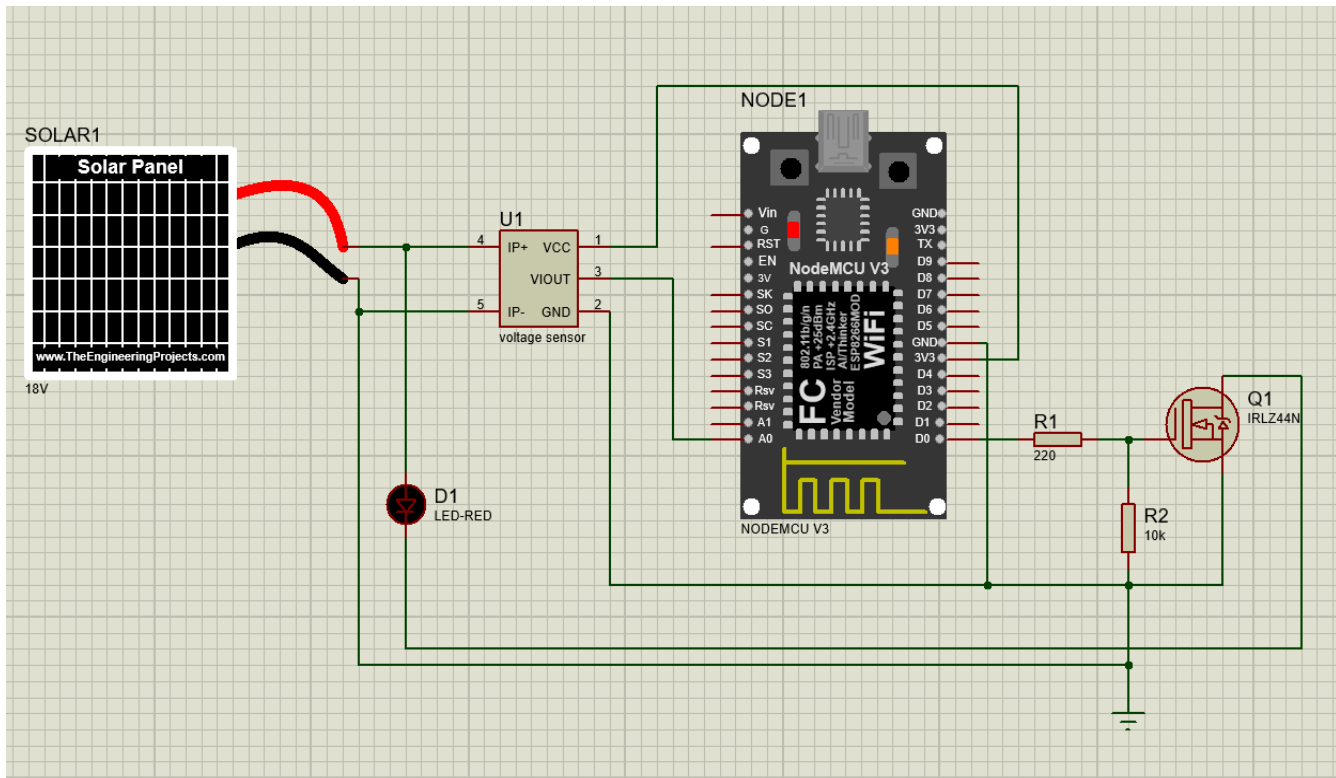


Fig 1. Circuit Diagram

2. Hardware Used:-

Sr. No.	Component Name	Hardware Specification	Quantity	Cost (Rs.)
1	Sensor	Voltage sensor module (25V)	1	90
2	Panel	Solar Panel (3W)	1	350
3	Microcontroller	ESP8266 (NodeMCU-V3)	2	499
4	Male-Female Jumper Wires	Standard breadboard jumper wires	4	30
5	Board	Bread Board / Zero PCB Board	1	110
6	Resistors	220 ohm and 10k ohm	2	10
7	USB / Data Cable	Micro USB cable for ESP32	1	80
8	LED	LED Strip (12V)	1	125
9	Sheet, Tape	White Sheet and 2-Way Tape	1	60
10	Strip	Male-Female Black Strip for Zeo PCB	4	60
11	MOSFET	IRLZ44N	1	40
Total Cost of Components				Rs. 1454

Table 1. Component Requirement for System

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2.1 Sensor:-

A voltage sensor module is used to measure the output voltage of the solar panel. It monitors the increase and decrease in voltage due to changes in sunlight intensity. The sensor sends the measured values to the microcontroller for processing, helping in analyzing the performance and efficiency of the solar panel system.

2.2 Solar Panel:-

We used a 3-watt solar panel in the project, which generates a maximum output voltage of around 16 to 18 volts under optimal sunlight conditions. The panel's voltage varies with changes in light intensity, and this variation is monitored using a voltage sensor module for analysis of the system's performance.

2.3 Microcontroller:-

The ESP8266 microcontroller is used in the project to process and manage the data obtained from the voltage sensor. It reads the measured voltage values and enables real-time monitoring of the system. The ESP8266 also provides built-in Wi-Fi capability, allowing data transmission and remote monitoring of the solar panel performance.

2.4 MOSFET:-

The IRLZ44N MOSFET is used in the project as a switching device to control the flow of power in the circuit. It operates efficiently with low gate voltage, making it suitable for use with microcontrollers like the ESP8266. The MOSFET helps in handling higher current loads and ensures reliable and fast switching performance in the system.

2.5 Register:-

We used 220 ohm and 10 k Ω resistors in the circuit for proper control and stability. The 220 ohm resistor is used to limit the current, protecting components like the MOSFET and microcontroller pins, while the 10 k Ω resistor is used as a pull-down resistor to ensure stable operation by preventing unwanted signals or floating conditions.

2.6 LED:-

We used a 12V LED in the project as the output load. It operates using the power supplied from the circuit and indicates the system's working status. The LED is controlled through the circuit components, allowing it to turn on or off based on the voltage and control signals.

3. Software Used:-

Sr. No.	Software Name	Purpose / Description
1	Arduino IDE	Used to write, compile, and upload code to Arduino boards; provides a simple interface for embedded system development.
2	Blynk IoT	Used for remote monitoring and control via mobile app; displays real-time data like voltage using widgets and provides notifications.
3	ESP32 Board Package	Enables the Arduino IDE to support ESP8266 microcontroller.
4	Serial Monitor	Used for debugging and viewing system output.

Table 2. Software Used for System

3.1 Blynk IoT:-

We used the Blynk IoT software in our project to enable smart monitoring and control of the system over the internet. Blynk provides a user-friendly mobile application and cloud platform where real-time data from the ESP8266, such as solar panel voltage, can be displayed using widgets like gauges, graphs, and value displays. The ESP8266 connects to the Blynk server via Wi-Fi and continuously sends sensor data for live monitoring. Blynk also allows users to control devices remotely, receive notifications, and store data for analysis. This makes the system more efficient, interactive, and accessible from anywhere using a smartphone or web dashboard.

3.2 Arduino IDE:-

The Arduino Integrated Development Environment (IDE) is a software application used to write, compile, and upload programs to Arduino boards such as the Arduino UNO and ESP32 Family. It provides a simple and user-friendly interface, making it easy for beginners and professionals to develop embedded system applications.

4. Implementation Procedure/Methodology:-

The system begins with the installation of a 3-watt solar panel, which acts as the primary source of energy. This solar panel converts sunlight into electrical energy. The output voltage of the panel varies depending on environmental conditions such as sunlight intensity, temperature, dust accumulation, and shading effects. Monitoring this variation is essential to determine the health and performance of the panel.

To measure the output voltage, a voltage sensor module is connected across the solar panel terminals. This sensor continuously senses the analog voltage generated by the panel and scales it down to a level that can be safely read by the microcontroller. The sensor ensures accurate and real-time voltage monitoring without damaging the controller.

The sensed analog voltage is then fed into the ESP8266 microcontroller, which is the main processing unit of the system. Since the ESP8266 operates digitally, it converts the incoming analog signal into digital data using its internal Analog-to-Digital Converter (ADC). The microcontroller is programmed using embedded C (Arduino IDE) to process this data efficiently.

Once the voltage data is received, the ESP8266 compares it with predefined threshold values stored in the program. These threshold values represent the normal operating range of the solar panel. If the voltage remains within this range, the system considers the panel to be functioning properly. However, if the voltage falls below or exceeds the set limits, it indicates abnormal conditions such as: Dust accumulation on the panel surface, Partial or complete shading, Panel degradation or fault, Loose connections or wiring issues.

When such abnormal conditions are detected, the ESP8266 triggers an alert mechanism. For local indication, an IRLZ44N MOSFET is used as an electronic switch. This MOSFET controls a 12V LED, which acts as a visual indicator. When a fault is detected, the MOSFET is turned ON, allowing current to flow and lighting up the LED. This provides an immediate on-site indication of system issues.

To ensure proper and safe operation of the circuit, supporting components are used. A 220-ohm resistor is connected in series with the LED to limit current and prevent damage. Additionally, a 10k-ohm pull-down resistor is connected to the MOSFET gate to ensure that it remains OFF when no signal is applied, thus avoiding false triggering and improving system stability.

For remote monitoring and smart functionality, the ESP8266 utilizes its built-in Wi-Fi capability to connect to the internet. It sends real-time voltage data to the Blynk IoT platform, which serves as the user interface. The Blynk application installed on a smartphone displays live voltage readings, system status, and graphical data trends.

In case of abnormal voltage conditions, the ESP8266 sends a notification through the Blynk platform. This alert is received on the user's smartphone, allowing them to take immediate corrective action. This feature eliminates the need for manual inspection and significantly reduces maintenance time.

Overall, this methodology ensures continuous monitoring, early fault detection, and real-time alerting. The integration of IoT technology improves the efficiency, reliability, and lifespan of the solar panel system while reducing operational and maintenance costs.

5. Block Diagram

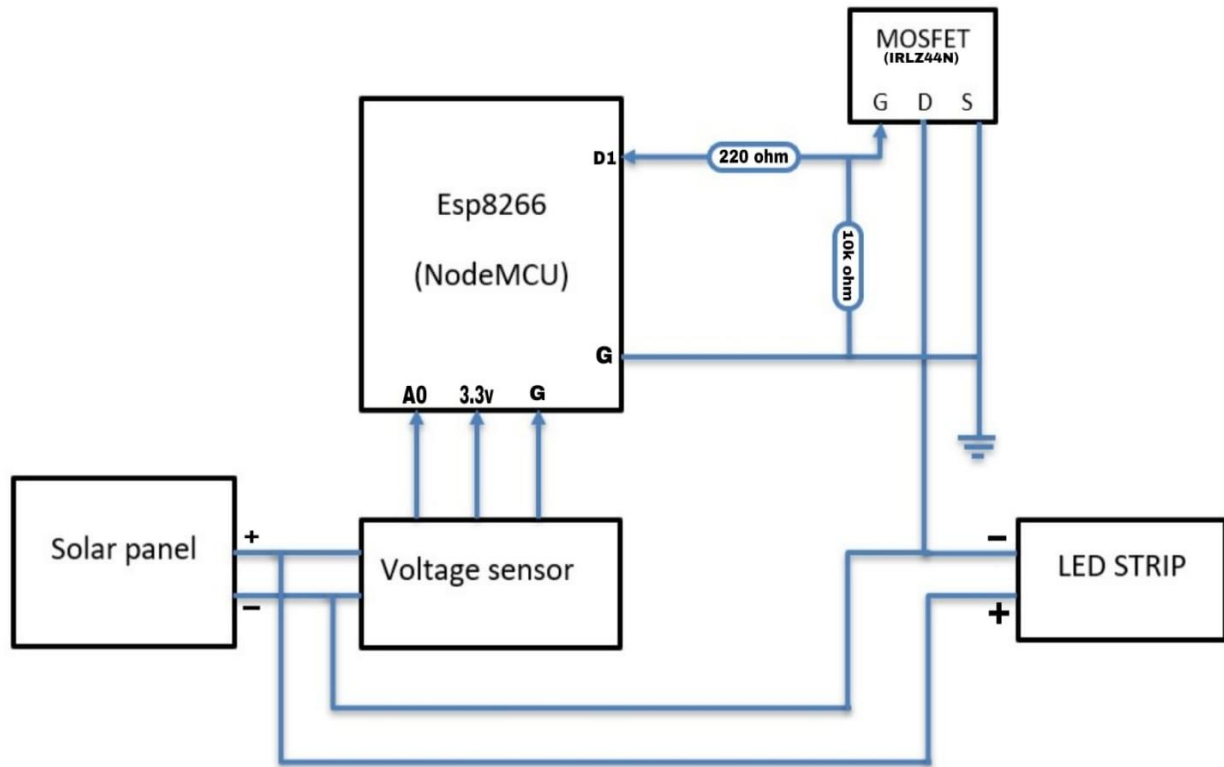


Fig 2. Block Diagram

6. Flowchart

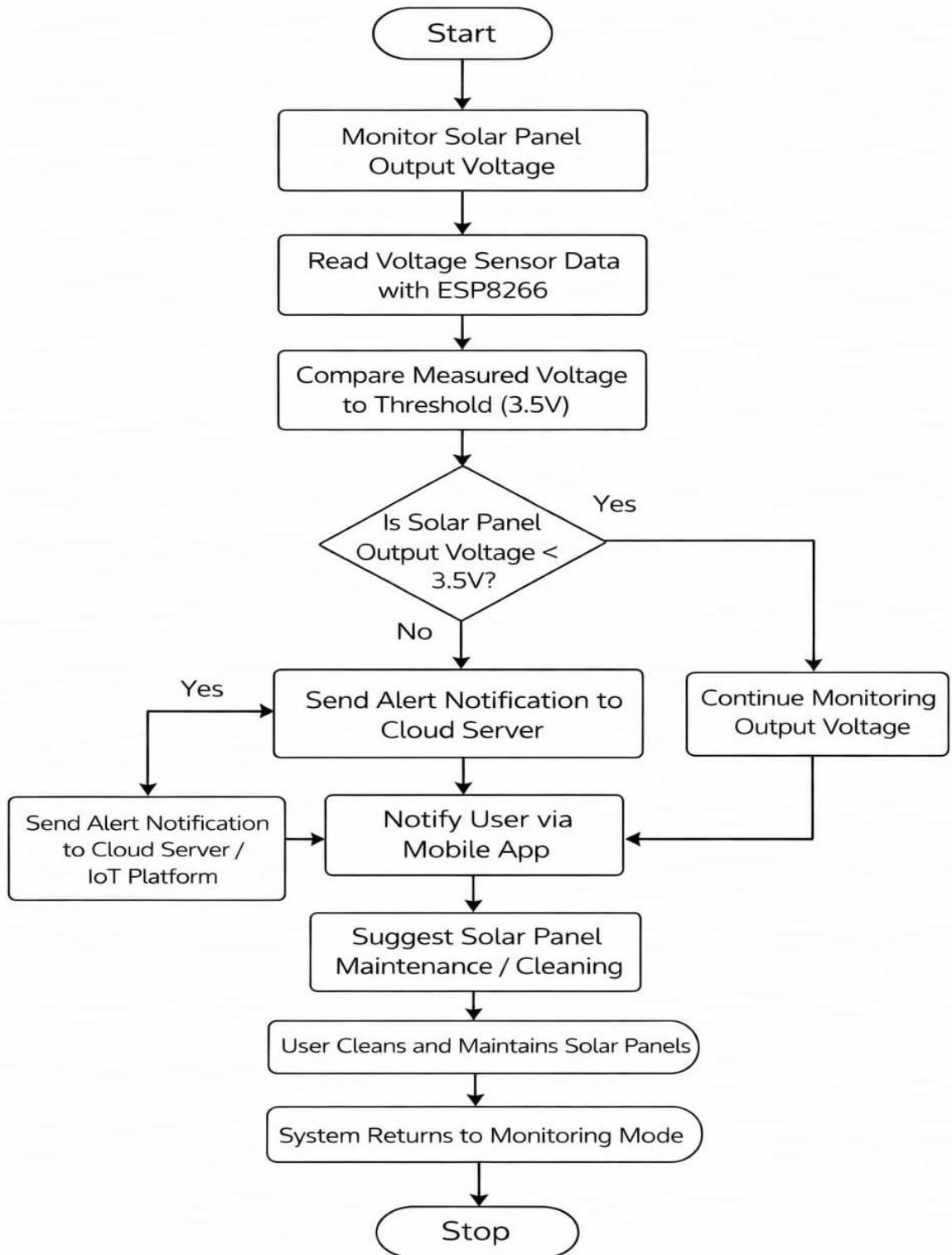


Fig 3. Flowchart

RESULT AND DISCUSSION

1. Hardware Results:-

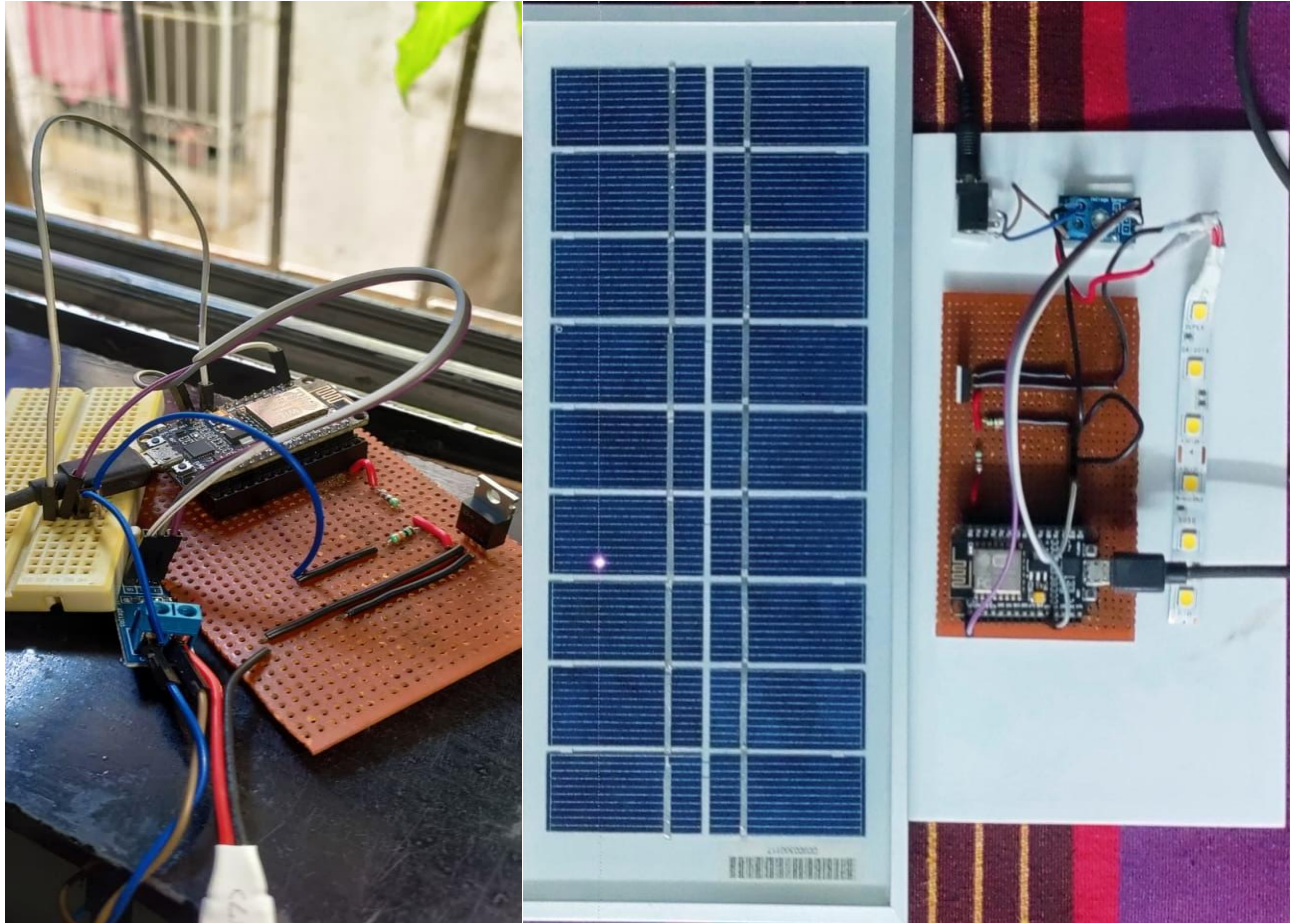


Fig 4. Hardware of Solar Panel Maintenance Alert System

2. Software Results:-

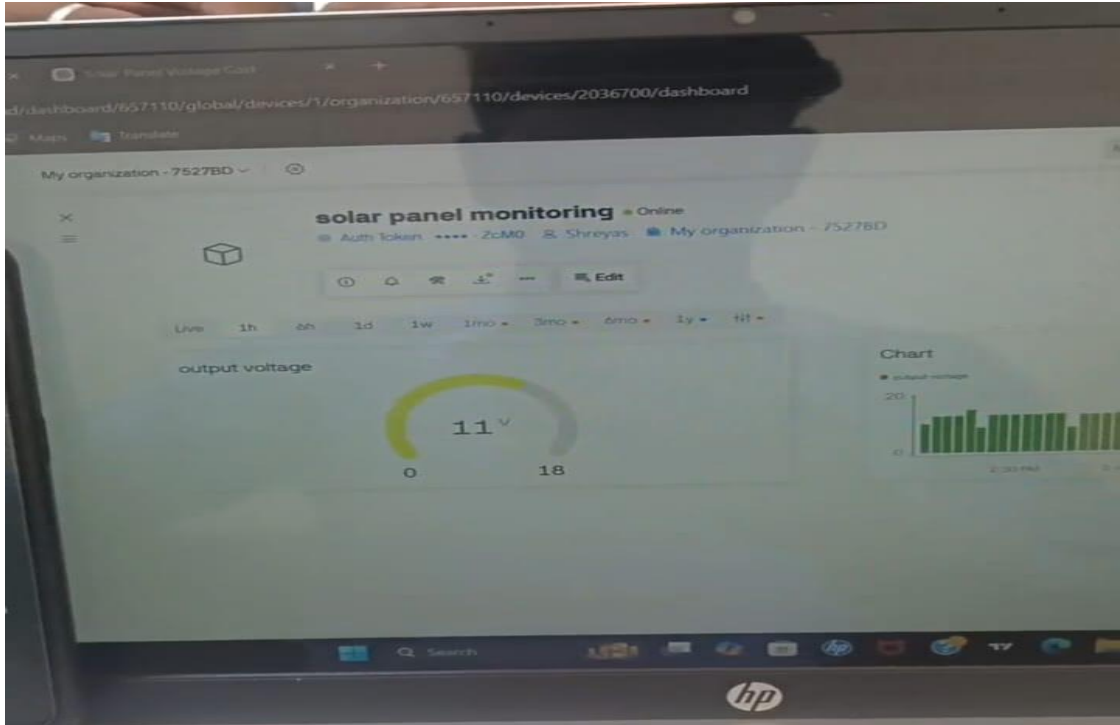


Fig 5. Output Show in BLYNK App

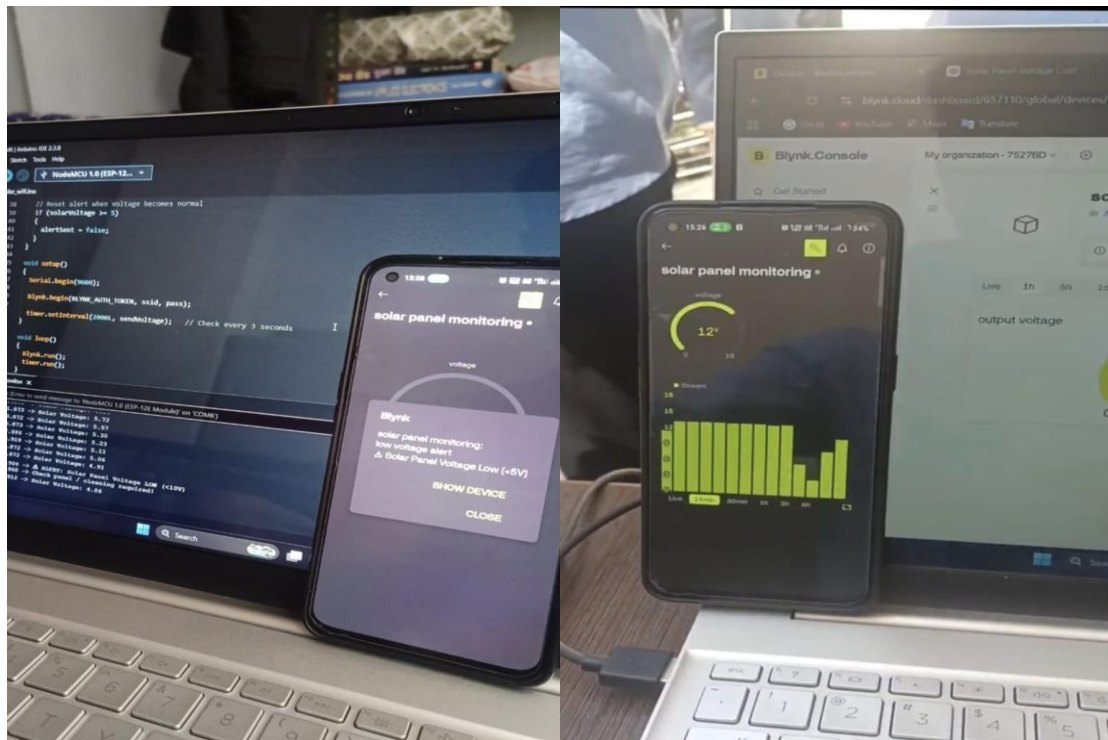


Fig 6. Message Alert via Mobile

CONCLUSION

The project “Solar Panel Maintenance Alert System” has been successfully designed and implemented to monitor the performance of a solar panel and provide timely alerts for maintenance. The system uses a voltage sensor to continuously measure the output of the solar panel and an ESP8266 microcontroller to process the data and compare it with a predefined threshold value.

Whenever the output voltage drops below the set limit, the system identifies it as a fault condition such as dust accumulation, shading, or improper connection and generates an alert through an LED indication or IoT-based notification. The use of a MOSFET allows efficient control of the load connected to the system.

This project reduces the need for manual inspection and helps in maintaining the efficiency of solar panels. It is cost-effective, simple to implement, and suitable for small-scale as well as real-time applications. Overall, the system demonstrates how embedded systems and IoT can be effectively used in renewable energy management to improve performance and reliability.

FUTURE SCOPE

The proposed solar panel monitoring and maintenance alert system has significant potential for further enhancement by incorporating advanced technologies and additional features. In its current form, the system focuses mainly on voltage monitoring; however, the inclusion of a current sensor along with the voltage sensor can improve the overall accuracy and provide better insights into power generation and system efficiency.

Furthermore, the system can be upgraded by integrating a dust sensor, which will directly detect dust accumulation on the surface of the solar panel. This will allow more precise identification of efficiency losses caused by environmental factors rather than relying only on voltage variations.

In the future, a fully automated cleaning mechanism can be developed and integrated with the system. This mechanism can automatically clean the solar panel whenever dust levels exceed a predefined threshold, thereby reducing manual effort and maintaining optimal performance of the panel.

The IoT functionality can also be enhanced by using advanced cloud platforms and mobile applications that provide features such as real-time data visualization, historical data analysis, performance reports, and remote system control. This will make the system more user-friendly and efficient for monitoring over long periods.

Additionally, the system can be scaled for large-scale solar power plants or solar farms by connecting multiple solar panels to a centralized monitoring system. This would allow operators to monitor and manage multiple units simultaneously, improving operational efficiency and fault management.

The integration of machine learning and artificial intelligence techniques can further enhance the system by enabling predictive maintenance. These technologies can analyze historical data to predict possible failures, optimize maintenance schedules, and improve the overall reliability of the system.

Moreover, future improvements may include:

- Integration with battery storage systems for complete energy management
- Use of GPS or location tracking for large installations
- Implementation of automatic fault diagnosis systems
- Development of a web-based dashboard for advanced monitoring

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